# Same history. Different write order. Different agent.

AGENT MEMORY

Your memory writer is a source of nondeterminism almost nobody tests.
8 slides - save for your next memory design review

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# Memory can go wrong while it is being written

THE RESEARCH

MemForm (arXiv, Oct 2026) studies memory formation: the step where a harness turns past runs into stored notes. Six LLMs tested. Existing defenses helped only partly.

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# Five interfaces, all exploitable in the paper's tests

WHERE FORMATION BREAKS

01 Content: what the note says
02 Evidence: the supporting context
03 Scope: where the note applies
04 Selection: which runs are kept
05 Path: the order they are consolidated in

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# Order alone can change what the agent does

THE SURPRISING ONE

In some setups, the same experiences consolidated in a different order led to different later behavior.

In production: two replicas, same history, drifting apart.

Lab result. Depends on the memory architecture.

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# We test the read path. The write path ships blind.

WHERE TESTS GO

READ PATH: Did retrieval find the right note? Gets evals and dashboards.

WRITE PATH: What was kept, in what order, saying what? Usually no tests at all.

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# Treat memory like a database migration

HOW I'D DESIGN IT

01 Raw episodes are append-only
02 Consolidation is a replayable job with a fixed order
03 Every stored fact is tagged with its source run
04 Diff memory after each consolidation
05 Rollback: drop one bad run, replay

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# The memory writer is part of your attack surface

RECAP

Formation can break at five interfaces
Consolidation order is a hidden variable
Replay, provenance and diffs make memory auditable
Cost: more storage, slower writes

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# How do you test your agent's memory writer today?

SAVE THIS

Save this for your next memory design review

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